Relative Boundary-Dissipation Certificates for Weak-Disorder Anderson Resolvents

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Abstract

We examine the Schur-recursion and law-stability interfaces in the claimed fixed-energy universality theorem for weak Anderson disorder on random regular graphs. A finite-block boundary replacement can be controlled in the metric of its actual imaginary self-energy, instead of multiplying an unweighted residual by a potentially large resolvent norm. We prove an explicit pseudohyperbolic error bound, sharp in a scalar example, together with two-sided covariance and retained log-determinant bounds. The theorem allows correlated boundary errors and cycles inside a finite block, subject to an invertible reference block and a complete relative-error certificate. Weighted row tests include discarded entries; a probabilistic good-event version supplies a disk-Wasserstein budget. We then keep the source's mean mode and multilevel fluctuation mode separate in the residual-to-law calculation. Worked examples explain why small ordinary second moments, an accurate finite solve, or a one-level iteration cannot replace boundary control. Original Ram perturbation proofs inform the error bookkeeping; Schur, dissipative, and hyperbolic identities retain their classical attribution. No finite-block statement establishes the infinite-tree boundary law or the finite-graph GOE replacement. Contribution assessment: C3/5, a source-specific relative boundary-error module with sharp disk and covariance bounds and an explicit law-residual interface, not an independent universality proof.

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1 Model, normalization, and logical scope

Fix an integer $d \geq 3$ and put $b = d - 1$. The source [1] uses ordinary adjacency, not adjacency divided by $\sqrt{b}$:

$$H_n = A_{n,d} + \operatorname{diag}(V_1, \ldots, V_n), \qquad V_i \overset{\text{iid}}{\sim} \operatorname{Unif}[-w, w].$$

The graph is uniform among simple labelled d-regular graphs, independently of the potentials. The resolvent convention is $G(z) = (H_n - z)^{-1}$ for $z = e + i\eta$, $\eta > 0$; its diagonal imaginary parts are positive. On the infinite regular tree let

$$\nu_{d,w}(B) = \mathbb{E}\langle \delta_o, \mathbf{1}_B(H_{\mathbb{T}_d})\delta_o \rangle.$$

The source asserts that, on any fixed compact clean-bulk interval, sufficiently small fixed $w > 0$ gives a continuous positive density $\rho_{d,w}$ and fixed-energy convergence of

$$\sum_i \delta_{n\rho_{d,w}(E)(\lambda_i - E)}$$

to the unit-intensity GOE bulk process, along all admissible sizes. Degree, disorder, energy, and compactly supported test are fixed before $n \to \infty$. The normalized GOE comparison has entry variances $1/n$ off the diagonal and $2/n$ on it, and its eigenvalues are scaled by n/π.

This report does not establish that global assertion. We prove deterministic finite-dimensional estimates and a conditional transfer of their error budgets to the source's law-stability inequalities. Tree existence, boundary inverse-margin moments, finite-graph switch estimates, conditional Gaussian rigidity, growing Gaussian insertion, and fixed-energy DBM remain separate inputs. Reading their proofs is not an independent certification of the entire chain.

All matrix norms below are Euclidean induced norms. For a complex matrix, $\operatorname{Im} S = (S - S^*)/(2i)$ is the Hermitian imaginary part. For a complex symmetric matrix it is the real symmetric matrix of entrywise imaginary parts. Transpose and adjoint are not interchangeable. Our readout vectors and boundary incidence matrices are real; the error matrices may be complex symmetric.

2 Source architecture and current context

The complete cached manuscript was read, including its scientific appendices. The following map records dependencies, not a claim to have independently reproved every imported result.

Source module	Mechanism and retained obligation
Tree	Disk coordinates; separated mean multiplier; multilevel one-leaf damping; inverse margins obtained by independence and absorption; common-environment boundary continuity.
Local law	Reversible marked switches, spectral averaging and sampled Ward bounds; two observations on one realized graph; boundary Schur replacement; descending-height bootstrap.
Marked entropy	Microscopic uniform integrability preserves second moments; conditional means require a separate potential perturbation; overlapping environment entropies cancel in a specified order.
Rigidity	Retained-coordinate rather than surface-volume normalization; environment-dependent precision shares; heat-flow equality and re-rooting identify Gaussian marks.
Resolvents	Spectral tail control leaves real constants; common-environment large-height limits fix them; root-excluded potential resampling preserves the law only unconditionally.
Insertion	Number of roots grows after taking the microscopic limit; outer and inner spectral cutoffs differ; signed rank-one sums are centered and replaced by GOE only at the smaller dimension.
DBM and counts	Exact free-convolution density adjustment; imported fixed-energy DBM; factorial multiplicities and finite Bonferroni brackets; reference GOE count moments close the brackets.

The boundary replacement targeted here is in `03-local-law.tex`, lines 382–395. The source controls the true finite Schur inverse on a high-probability event and uses an ordinary inverse perturbation bound. The new certificate is an alternative *when the stronger relative boundary hypothesis is available*. It does not invalidate that source argument, nor assert an improved random-graph rate without a new probabilistic estimate of the relative error.

The source already proves the retained-star identity $\det K_o^{\mathrm{ret}} = |m_o|^2 \prod_i \Im g_i$ and its volume convention in `05-rigidity.tex`, lines 37–115. We retain and use that identity; we do not count its rederivation as a contribution.

2.1 Primary literature checked on October 9, 2026

Froese, Hasler, and Spitzer [2] give a geometric proof of weak-disorder absolute continuity on the degree-three tree. Their primary statement requires a small weighted fourth-moment tail; their recursion and hyperbolic-distance formulation make clear that this is not a finite-graph eigenvalue-statistics theorem. We checked the model, main statements and the displayed geometric/moment framework, not their entire proof.

The new preprint of He, Huang, and Wang, submitted October 6, 2026 [3], claims fixed-energy bulk universality for the *clean* uniform simple regular graph at every fixed $d \geq 3$. Its displayed main theorem uses $n\pi\rho_d(E)$ scaling and a Sine_1 process of intensity $1/\pi$; dividing those coordinates by π gives the unit-intensity convention used above. The checked introduction attributes the mechanism to mixed fourth moments and microscopic switching loop equations. This is a current preprint claim, not an independently audited proof or a weak-disorder theorem.

The dissipative inverse identity, Schur complement, Woodbury formula, weighted Schur test, and invariance of the half-plane metric are classical. Our incremental result is their explicit combination for the actual boundary replacement, including a sharp relative threshold, covariance loss, and probabilistic residual interface. No historical priority claim for a general perturbation principle is made.

3 What the tree recursion does and does not contract

On a forward tree, deletion of the parent edge gives

$$g = (V - e - i\eta - \textstyle\sum_{i=1}^{b} g_i)^{-1}. \tag{1}$$

The child branches are independent under the actual tree law. For e in a fixed compact $J \Subset (-2\sqrt{b}, 2\sqrt{b})$ set

$$q(e) = \frac{-e + i\sqrt{4b - e^2}}{2b}, \qquad X_e(g) = \frac{g - q}{g - \bar{q}}, \qquad T_e(x) = \frac{q - x\bar{q}}{1 - x}.$$

These maps identify $\mathbb{H}$ with $\mathbb{D}$. In particular

$$1 - |X_e(g)|^2 = \frac{4\Im q\, \Im g}{|g - \bar{q}|^2}, \qquad |T_e(x) - T_e(y)| = \frac{2\Im q\, |x - y|}{|1 - x||1 - y|}. \tag{2}$$

Thus disk transportation does not by itself control unweighted half-plane errors. The source separately proves moments of $D(x) = (1 - |x|)^{-1}$.

At zero disorder and height, a perturbation of just one child has map

$$H(x) = \frac{ax}{b - (b-1)x}, \qquad a = bq^2, \qquad |a| = 1.$$

After j levels, with all other inputs clean,

$$H^{\circ j}(x) = \frac{(a/b)^j x}{1 - c_j x}, \qquad c_j = (1 - b^{-1}) \sum_{k=0}^{j-1} (a/b)^k. \tag{3}$$

Composition proves the formula by the recurrence $c_{j+1} = c_j + (1 - b^{-1})(a/b)^j$. Its limit $(b-1)/(b-a)$ has modulus strictly below one on J, because $a \neq 1$ there. Hence one-leaf derivatives are $O_J(b^{-j})$ for large fixed j. However the sum of the b one-level derivatives at the clean input is a, of modulus one. There is no strict one-step contraction of the mean. Replacing this phase by its absolute value loses the separate invertibility of $1 - a$.

Example 3.1 (A small disk norm with an uncontrolled boundary tail)**.** For $0 < \epsilon < 1/2$, let $X_\epsilon = 1 - \epsilon^3$ with probability ϵ, and zero otherwise. Then $\mathbb{E}|X_\epsilon|^2 \le \epsilon$, whereas $\mathbb{E}D(X_\epsilon)^p \ge \epsilon^{1-3p}$. At $e = 0$, $q = i/\sqrt{b}$ and $|T_e(1 - \epsilon^3)| = (2 - \epsilon^3)/(\sqrt{b}\,\epsilon^3)$. Thus even $\mathbb{E}|T_e(X_\epsilon)|^2$ diverges as $\epsilon \downarrow 0$. This is a manufactured law, not the actual tree law; it explains why the source's additional inverse-margin argument is essential.

There is also a purely deterministic pole mechanism. Two bounded half-plane inputs $g_1 = 1 + i\delta$, $g_2 = -1 + i\delta$, with $V = e = \eta = 0$, give $g = i/(2\delta)$. Bounded child magnitudes do not provide a lower bound on the imaginary denominator. Our certificate below measures error against the *actual* imaginary part, rather than assuming it away.

4 A relative boundary-dissipation theorem

Let H, D be real symmetric $N \times N$ matrices, $D \ge 0$, and suppose

$$S_0 = H - iD \quad \text{is invertible}, \qquad G_0 = S_0^{-1}.$$

Let $C \in \mathbb{R}^{N\times r}$ satisfy $CC^{\mathsf{T}} \le D$. It need not be square or injective. Consider

$$S_1 = S_0 - CRC^{\mathsf{T}}, \qquad R = R^{\mathsf{T}} \in \mathbb{C}^{r\times r}, \qquad \|R\| \le \tau < 1. \tag{4}$$

The matrix R measures error in the boundary dissipation metric. There is no assumption that $\|G_0\| \left\|CRC^{\mathsf{T}}\right\| < 1$.

Lemma 4.1 (Contractive boundary compression)**.** *The matrix $K = C^{\mathsf{T}}G_0C$ has $\|K\| \le 1$. Moreover, S_1 is invertible, and*

$$G_1 = G_0 + G_0CR(I - KR)^{-1}C^{\mathsf{T}}G_0. \tag{5}$$

Proof. The exact inverse identity gives

$$\operatorname{Im} G_0 = G_0^* D G_0. \tag{6}$$

For $u \in \mathbb{C}^r$, put $x = G_0Cu$. Then

$$\|Ku\|^2 = \left\|C^{\mathsf{T}}x\right\|^2 \le x^*Dx = \Im(u^*Ku) \le \|u\|\ \|Ku\|.$$

This proves contractivity. Since $\|KR\| < 1$, the middle inverse exists. Direct multiplication of (5) by $S_0 - CRC^{\mathsf{T}}$ gives the identity. Equivalently the finite determinant identity proves nonsingularity. The order of K and R has not been commuted. □

Theorem 4.2 (Sharp readout and covariance bounds)**.** *Under (4), write $G_1 = S_1^{-1}$. For every real v with $y_0 := \Im(v^{\mathsf{T}}G_0v) > 0$, put $g_j = v^{\mathsf{T}}G_jv$ and $y_j = \Im g_j$. Then $y_1 > 0$ and*

$$\frac{|g_1 - g_0|^2}{y_0y_1} \le \frac{\tau^2}{1-\tau}, \tag{7}$$

$$\frac{|g_1 - g_0|}{|g_1 - \bar{g}_0|} \le \frac{\tau}{2-\tau}, \tag{8}$$

$$(1-\tau)y_0 \le y_1 \le (1-\tau)^{-1}y_0, \qquad |g_1 - g_0| \le \frac{\tau}{1-\tau}y_0. \tag{9}$$

In real symmetric matrix order,

$$(1-\tau)\operatorname{Im} G_0 \le \operatorname{Im} G_1 \le (1-\tau)^{-1}\operatorname{Im} G_0. \tag{10}$$

If a real $N \times m$ observation matrix L has $K_0 = L^{\mathsf{T}}(\operatorname{Im} G_0)L > 0$, then $K_1 > 0$ and

$$|\log\det K_1 - \log\det K_0| \le -m\log(1-\tau). \tag{11}$$

Proof. Set $D_1 = -\operatorname{Im} S_1 = D + C(\operatorname{Im} R)C^{\mathsf{T}}$. Since $\|\operatorname{Im} R\| \le \tau$,

$$D_1 \ge D - \tau CC^{\mathsf{T}} \ge (1-\tau)D \ge (1-\tau)CC^{\mathsf{T}}.$$

Consequently $\operatorname{Im} G_1 = G_1^*D_1G_1 \ge 0$. The inverse identity, using symmetry and the reality of v, yields

$$g_1 - g_0 = (C^{\mathsf{T}}G_0v)^{\mathsf{T}}R(C^{\mathsf{T}}G_1v).$$

Complex bilinear Cauchy–Schwarz and the two energy identities give

$$|g_1 - g_0| \le \tau\sqrt{y_0}\sqrt{y_1/(1-\tau)}. \tag{12}$$

If $y_1 = 0$, this forces $g_1 = g_0$ and hence $y_0 = 0$, a contradiction. Division proves (7). The identity $|g_1 - \bar{g}_0|^2 = |g_1 - g_0|^2 + 4y_0y_1$ gives (8), since $\tau^2 + 4(1-\tau) = (2-\tau)^2$.

Writing $s = y_1/y_0$, the imaginary part alone gives $(s-1)^2/s \le \tau^2/(1-\tau)$. The two roots of the associated quadratic are $1-\tau$ and $(1-\tau)^{-1}$, proving the ratio bounds. Their upper bound in (12) gives the absolute bound. When $y_0 = 0$, the same bilinear identity forces $g_1 = g_0$ and $y_1 = 0$; thus the ratio inequalities extend as quadratic-form inequalities. Because both imaginary matrices are real symmetric, testing all real v proves (10).

Congruence by L, then by $K_0^{-1/2}$, places every eigenvalue of $K_0^{-1/2}K_1K_0^{-1/2}$ in $[1-\tau, (1-\tau)^{-1}]$. Summing their logarithms proves (11). No assertion that the two covariances commute is used. □

For centered Gaussian observations in these same m real coordinates, the differential-entropy difference is at most $-\frac{m}{2}\log(1-\tau)$ in absolute value. This controls the *reference covariance* contribution, not the entropy of arbitrary non-Gaussian marks or the heat-flow equality case.

Example 4.3 (Sharpness)**.** Take $S_0 = -i$, $C = 1$, and $R = -i\tau$. Then $G_0 = i$, $G_1 = i/(1-\tau)$, attaining the hyperbolic, pseudohyperbolic, upper-ratio and absolute-error bounds. At $\tau = 1$ the updated matrix is singular. The strict threshold cannot be dropped for arbitrary complex symmetric errors. Real errors are a different subclass and need not cause this particular loss of dissipation.

5 The actual finite-tree boundary replacement

Let $\mathcal{T}$ be the finite internal tree used for a depth-j recursion. Index its external child ports by $1, \dots, r$. The real incidence matrix W has one column e_p for each port attached to internal parent p; several ports may share that parent. The reference independent cavity values give

$$\Gamma_0 = \operatorname{diag}(g_1, \dots, g_r), \qquad \Im g_i > 0. \quad S_0 = A_{\mathcal{T}} + \operatorname{diag}(V) - eI - i\eta I - W\Gamma_0 W^{\mathsf{T}}.$$

The sign before the boundary self-energy is negative. The leaf-elimination formula for $(S_0^{-1})_{oo}$ is exactly (1). An actual correlated exterior compression $\Gamma = \Gamma_0 + \Delta$ gives $S_1 = S_0 - W\Delta W^{\mathsf{T}}$, without requiring Δ to be diagonal. Writing

$$B = \operatorname{Im}\Gamma_0 = \operatorname{diag}(\Im g_i), \quad C = WB^{1/2}, \quad R = B^{-1/2}\Delta B^{-1/2}, \tag{13}$$

we have $-\operatorname{Im} S_0 = \eta I + CC^{\mathsf{T}}$. Theorem 4.2 therefore applies whenever the complete $\|R\| \le \tau < 1$ is certified. At positive height the actual exterior resolvent is complex symmetric. The theorem also allows an approximate complex symmetric exterior whose admissibility follows from the displayed relative bound.

Lemma 5.1 (Zero-height reference branch)**.** *At $\eta = 0$, the reference block just described is invertible for a full finite forward tree with $b \ge 2$ children per nonterminal vertex, including the full-root variant with $d \ge 3$ root children, if every outer internal vertex has a port with strictly positive imaginary message. Every diagonal entry of S_0^{-1} then has positive imaginary part.*

Proof. If $S_0u = 0$, the imaginary part of u^*S_0u forces $u = 0$ on the outer internal level. The equation at an outer vertex then forces its parent value to vanish: every other value in that row is already zero and the parent edge has coefficient one. Repeating level by level gives $u = 0$ throughout. Thus S_0 is invertible.

For a real source $v = e_x$, if the corresponding diagonal imaginary part were zero, the energy identity would make $u = S_0^{-1}e_x$ vanish at the outer level. At any level being peeled, at most one row has nonzero right-hand side, namely the x row. Each parent has at least two children, so one of

its child rows is homogeneous. Since that child's value and all its children's values already vanish, this row forces the parent value to vanish. Induction makes every value zero, after which the x row reads $0 = 1$. For the one-vertex block positivity is immediate. The branching hypothesis matters: on a two-vertex path with $S_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -i \end{smallmatrix}\right)$, the second diagonal of S_0^{-1} is zero despite invertibility. □

For a general real symmetric finite block, including one with cycles, invertibility of S_0 and positivity of the chosen readout remain explicit requirements; they do not follow from the presence of one damped vertex. For example a three-vertex path damped only at its middle vertex has the undamped zero vector $(1, 0, -1)$ at energy zero. The algebraic theorem allows cycles but supplies no missing observability condition.

5.1 Disk errors and retained Gaussian coordinates

The pseudohyperbolic distance

$$p_{\mathbb{H}}(g, h) = \frac{|g - h|}{|g - \bar{h}|}$$

equals $|X_e(g) - X_e(h)|/|1 - \overline{X_e(h)} X_e(g)|$. This follows by substituting the two fractional-linear formulas; all common factors cancel. Since the disk denominator is at most two,

$$|X_e(g_1) - X_e(g_0)| \le \frac{2\tau}{2 - \tau}. \tag{14}$$

The bound has no explicit inverse power of η or factor depending on tree depth. This does *not* make its hypothesis uniform in depth: the number of ports and their smallest imaginary parts may deteriorate. The physical absolute error in (9) still scales with $\Im g_0$, which itself need not be bounded.

For the source star one can take the real coordinate selector L which omits one neighbor. The source proves $K_0 > 0$ in these coordinates. Thus (11) gives an explicit error budget for that retained log-determinant, without replacing the constraint volume by Euclidean surface volume. Tensoring m independent colors multiplies the log-determinant error by m; it does not create independence of actual eigenvector colors or establish their conditional Gaussian law.

Example 5.2 (Why the ordinary inverse-norm test may fail)**.** For $0 < t \le 1$ and $0 < \tau < 1$, consider the manufactured weighted two-vertex block

$$S_0 = \begin{pmatrix} 0 & t \\ t & -i \end{pmatrix}, \quad C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad R = -i\tau.$$

Then

$$G_0 = \begin{pmatrix} i/t^2 & 1/t \\ 1/t & 0 \end{pmatrix}, \qquad (G_1)_{11} = i(1 - \tau)/t^2.$$

The scalar hyperbolic bound is again attained, now at the lower imaginary-ratio endpoint. Yet $\|G_0\| \left\|CRC^{\mathsf{T}}\right\| \ge \tau/t^2$, which can exceed one arbitrarily badly. The ordinary sufficient inverse-norm test then says nothing while the relative certificate remains valid. This is a finite weighted-block example, not a realization or simulation of the ordinary random regular graph model. The gain is in certification, not a claimed improvement of the disorder threshold.

6 Complete residual tests in the boundary metric

Forming Δ numerically is not enough: it must include the difference between the true exterior compression and its chosen approximation, including all retained and discarded entries. Suppose $B = \mathrm{diag}(b_i)$ with certified $b_i \geq \underline{b}_i > 0$ and $|\Delta_{ij}| \leq e_{ij} = e_{ji} \geq 0$. The bounds concern complex moduli, not just real parts.

Proposition 6.1 (Rational weighted-row certificate)**.** *If positive numbers s_i and $0 \leq \tau < 1$ satisfy*

$$\sum_j e_{ij} s_j \leq \tau\, \underline{b}_i\, s_i \qquad \text{for every } i, \tag{15}$$

then $\left\| B^{-1/2} \Delta B^{-1/2} \right\| \leq \tau$. In particular rational upper and lower bounds in (15) *give an exact finite arithmetic certificate.*

Proof. Put $A_{ij} = e_{ij}/\sqrt{b_i b_j}$ and $v_i = \sqrt{b_i} s_i$. Then $Av \leq \tau v$. For real nonnegative z,

$$\sum_{ij} A_{ij} z_i z_j \leq \frac{1}{2} \sum_{ij} A_{ij} \left(z_i^2 \frac{v_j}{v_i} + z_j^2 \frac{v_i}{v_j} \right) = \sum_i \frac{(Av)_i}{v_i} z_i^2 \leq \tau \, \|z\|^2 .$$

The same domination applied to absolute values bounds the induced norm of the symmetric nonnegative matrix A by τ. For complex vectors, entrywise modulus comparison gives $|B^{-1/2} \Delta B^{-1/2} u| \leq A|u|$. This proves the assertion without computing square roots in the certificate inequalities. □

For an implemented approximation $\widehat{\Delta}$, let e_{ij}^{ret} bound its entries, e_{ij}^{tail} bound the uncalculated/model entries, and e_{ij}^{alg} bound arithmetic and assembly errors. The sufficient test is (15) with $e = e^{\mathrm{ret}} + e^{\mathrm{tail}} + e^{\mathrm{alg}}$. A zero written into an uncomputed entry is not a tail bound. The weights may be optimized on the finite instance, but their existence and all row inequalities must be certified; there is no claimed general complexity improvement.

Example 6.2 (An anisotropic two-port budget)**.** Let $\underline{b} = (1, 4)$ and suppose $e_{11} = 1/10$, $e_{22} = 2/5$, $e_{12} = e_{21} = 1/5$. With $s = (1, 1/2)$, the two row quotients in (15) are both $1/5$. Thus $\tau = 1/5$, the root disk error is at most $2/9$, the imaginary ratio lies in $[4/5, 5/4]$, and an m-coordinate retained log-determinant error is at most $m \log(5/4)$. The unit-weight test would instead return $3/10$. The improvement is for this explicit metric budget, not a new random-matrix estimate.

If $\eta > 0$, an arbitrary internal error can formally be placed in the full dissipation metric $D^{-1/2} E D^{-1/2}$. At height zero D may be singular, so this expression need not exist. Internal errors outside the boundary-supported factorization are not covered by (13); they need a separate inverse/residual certificate or a proved extension of the metric. This limitation is especially relevant to approximate internal solves.

7 From block errors to a conditional law certificate

We first formulate a finite probabilistic consequence which makes no independence assumption. Couple an exact block and its approximation on one probability space with a common disk coordinate X_e. Suppose on an event $\mathcal{A}$ they satisfy Theorem 4.2, with a measurable $0 \leq \tau < 1$. Outside $\mathcal{A}$ assume only that both root outputs lie in $\mathbb{D}$. Writing these outputs as $U, \widehat{U}$, the coupling gives

$$W_2(\mathrm{Law}(U), \mathrm{Law}(\widehat{U}))^2 \leq \mathbb{E}\left[\mathbf{1}_{\mathcal{A}} \left(\frac{2\tau}{2 - \tau} \right)^2 \right] + 4\mathbb{P}(\mathcal{A}^c). \tag{16}$$

Indeed (14) holds on the good event and the disk diameter is two elsewhere. An event with small probability can still have a large half-plane error; (16) claims only the specified bounded-coordinate Wasserstein control.

7.1 Keeping the two stability modes separate

Let $\Phi_j(P)$ be the source's depth-j law map with iid leaves of law P and independent internal potentials. Let Q be the actual tree disk law. The next result takes as hypotheses the exact coupling inequalities produced in the source, not merely numerical derivatives of Φ_j.

Proposition 7.1 (Two-residual stability budget)**.** *Let (X, Z) be an optimal quadratic coupling of P, Q, and put $B_0 = \|X - Z\|_{L^2}$ and $M = \mathbb{E}(X - Z)$. Assume $Q = \Phi_j(Q)$ for $j = 1, \ell$, and assume coupling the leaf pairs independently with common internal potentials gives*

$$|\mathbb{E}(F_1(X_\cdot) - F_1(Z_\cdot)) - aM| \le \epsilon_1 B_0, \tag{17}$$

$$\|F_\ell(X_\cdot) - F_\ell(Z_\cdot)\|_{L^2} \le \kappa B_0 + A|M|. \tag{18}$$

Suppose $s = |1 - a| > 0$, $u_j = W_2(P, \Phi_j(P))$, and $\delta = 1 - \kappa - A\epsilon_1/s > 0$. Then

$$W_2(P, Q) \le \frac{u_\ell + (A/s)u_1}{\delta}. \tag{19}$$

If implemented maps $\widehat{\Phi}_j$ satisfy $W_2(\widehat{\Phi}_j(P), \Phi_j(P)) \le e_j$, one may replace u_j by the certified sum $W_2(P, \widehat{\Phi}_j(P)) + e_j$, with e_j supplied by (16) *when its block premises hold.*

Proof. The expectation difference between P and $\Phi_1(P)$ is at most u_1, because expectation is one-Lipschitz for W_2. Stationarity and (17) therefore give $s|M| \le u_1 + \epsilon_1 B_0$. The triangle inequality and the displayed depth-ℓ coupling give $B_0 \le u_\ell + \kappa B_0 + A|M|$. Substitution and division by the strictly positive δ prove (19). The last statement is another triangle inequality, not a claim that the implemented recursion has a fixed point. □

The source obtains these coupling hypotheses only for small marginal disk norms, a fixed large depth ℓ, sufficiently small w, η, and sufficiently high inverse-margin moments for the reference Q. The arbitrary comparison law P has no such margin assumption. The key sibling bound excludes the varied leaf from its margin coefficient. Replacing that argument by a global derivative supremum would lose this asymmetric feature. We import those hypotheses and constants; the proposition makes their final residual arithmetic explicit.

For example, $\kappa = 1/5$, $A = 2$, $s = 1$ and $\epsilon_1 = 1/20$ give $\delta = 7/10$ and $B_0 \le (10/7)(u_\ell + 2u_1)$. These are manufactured admissible constants, not evaluated constants for the source tree law. The factor A/s also displays the loss when the mean multiplier approaches one near a spectral edge.

7.2 What the combination does not give

The relative block certificate requires a coupled *full* boundary error, not independent sampling of two unrelated environments. The source's conditional empirical recursion needs independence of leaf samples given the realized graph and a two-observation comparison on that same graph. Neither property follows from (16). Likewise, verifying finitely many levels of a truncated recursion does not show that their boundary law is the infinite-tree law.

To turn the module into a quantitative local-law improvement one would have to prove, at the target height, a bound on the probability that $B^{-1/2}\Delta B^{-1/2}$ fails its threshold, together with its truncated second moment on the good event. Small entrywise absolute errors cannot be divided by unproved lower bounds on B. This is precisely where the source's inverse-margin and spectral-averaging inputs remain substantive.

8 Original proof support, costs, and extensions

Fresh reading of the original stability text [4] includes its finite inverse perturbation proof, its nonlinear residual ball theorem, the radii inequality, and its exact resolvent identity. The finite inverse proof explains why a residual must be paired with an actual lower stability margin in the declared norm. The residual-ball proof requires a genuine contraction on an admissible ball and an injective observation isomorphism; it cannot be applied to a law map whose mean derivative has modulus one. Its perturbation conclusion also requires the perturbed root to lie in that ball.

Our result uses a different, problem-specific substitute: the boundary compression is contractive by dissipation, even if the full resolvent is large. This is not a claim that the original generic theorem proves tree-law existence. At zero height the boundary metric may be singular in internal directions, so it is deliberately not presented as a bounded isomorphism of all state coordinates.

The freshly read later core5 comparison [5] supplies the ordered Woodbury and inexact Schur-residual identities. It correctly distinguishes a reduced interface residual from the full residual: eliminated forcing and interior solve errors must be retained. We use it as a later comparison, not as the original source of classical Schur algebra. The exact independent proofs in this report do not require a claim of exhaustive private-corpus reading.

8.1 Advantages and limitations

The main advantage is an explicit bound on a physically relevant readout and covariance without an unweighted full inverse norm. Correlated boundary errors are allowed, the constants are independent of the block dimension once its relative budget is fixed, and the retained-coordinate log-determinant budget is immediate. The sharp examples distinguish this from simply renaming an ordinary Neumann inequality.

The cost is stronger information about the boundary self-energy. Constructing a rigorous lower imaginary margin and a complete matrix error may be more expensive than solving the original finite block. A depth-j forward tree has b^j ports; dense boundary assembly can be quadratic and dense factorization cubic in that port count. Sparsity or a verified low-rank representation may help, but no timing, asymptotic speedup, or computed $w_0(d, \kappa)$ is claimed.

Covariance control does not prove entropy rigidity. A well-conditioned reference Gaussian says nothing by itself about the equality case of the non-Gaussian heat flow. Nor does a root resolvent certificate identify microscopic point processes. The source's real constants, resampling invariance, signed rank-one replacement, growing-cutoff order, DBM density normalization and reference count moments remain untouched.

8.2 Feasible extensions

One direction is an environment-dependent port partition: certify relative errors blockwise, retaining cross-block remainders in a weighted row test. A second is validated interval arithmetic for the imaginary port lower bounds, combined with a probabilistic budget for rejected environments. A third is an observable-specific refinement which uses the actual boundary response vectors in (12) rather than their energy upper bounds. Each requires complete tail/rounding estimates; none is implied by an apparently converged population-dynamics sample.

At a changing energy, the clean coordinate $q(e)$ also changes. This report's coupling formulas use one common X_e; comparison across energies must add that coordinate-change error and retain the source's same-environment continuity argument. Uniform control up to a spectral edge or as $w \downarrow 0$ is not asserted.

9 Read scope and contribution assessment

The owned ledger records hashes captured before fresh literal displays, the exact displayed intervals, unread complements, and unchanged post-read hashes. All 12 cached public texts were read in full: 4,612 lines, including the bibliography and README. This does not mean that every cited external or companion theorem was read in full or independently proved. The original private stability file was read at lines 1–295; the later core5 comparison at lines 189–248 and 364–436. All remaining private corpus is outside the claimed read scope.

Independent executable checks accompany the report. They test finite rational complex matrix fixtures, the exact boundary inverse and dissipation identities, relative covariance and scalar bounds, sharp examples, weighted-row budgets, and the two-mode arithmetic. Their 300 fixture cases and 3,007 assertions are recorded separately in the owned receipt. They are not random-graph experiments, infinite-law certificates, or replays of foreign checkers. Parent checks are not included.

No mathematical correction to the source is asserted on this scope. The one-leaf mechanism, asymmetric inverse-margin requirement, retained star determinant, conditioned environments, and unconditional switch invariance have all been preserved. The current-literature update is explicitly versioned and distinguishes clean from disordered models.

Applicability remains L3. C3 is assigned to the demonstrated local increment in Theorem 4.2 and its actual Schur/law interfaces, not to the textbook identities separately. It is not C4: no central probabilistic estimate or universality theorem has been replaced or independently established, and the complete relative error is still an input that must be certified in an application.

References

[1] OpenAI, *Fixed-energy universality for weak Anderson disorder on random regular graphs*, October 5, 2026, pinned public manuscript. Public manuscript and source directory. Cached text authenticated against the captured October 8 tree; not live branch verification.

[2] R. Froese, D. Hasler and W. Spitzer, *Absolutely continuous spectrum for the Anderson model on a tree: a geometric proof of Klein's theorem*, Communications in Mathematical Physics 269 (2007), 239–257. `https://arxiv.org/abs/math-ph/0511050`. Primary model/statements and geometric framework checked; no full proof audit.

[3] Y. He, J. Huang and X. Wang, *Bulk universality of random regular graphs*, arXiv:2610.08127v1, October 6, 2026. `https://arxiv.org/abs/2610.08127`. Primary abstract, model and main statements checked; recent clean-model preprint claim, not independently verified.

[4] M. Subbarao, original author-supplied private TR1, *Perturbation, regularization, and a posteriori control*, `stability.tex`, lines 1–295. Confidential local proof text; not an online source.

[5] M. Subbarao, author-supplied private core5 report, lines 189–248 and 364–436: norm-controlled inverse tails, Schur reconstruction residuals and low-rank reuse. Later comparison, not a substitute for the original proof text or a novelty claim for classical matrix identities.